



Ontario Future Labour Force Needs Study

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An aerial photograph of a winding asphalt road that snakes through a lush, green valley. The road has white dashed lines and a solid white edge. The surrounding landscape is densely forested with various shades of green. The overall image has a blue tint.

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1. Executive Summary

The Council of Ontario Universities commissioned research by Stokes Economics (Stokes) to provide occupation demand projections for university educated workers over the period of 2021 to 2030. There are 122 occupations under the skill level A classification within the National Occupational Classification system where a university degree or certificate is usually required.

The Stokes Economics July 2021 forecast provides the economic backdrop for the employment and occupation forecasts used in this report.

Stokes took a modelled approach to produce the occupational projections. The models relied on data from the 2016 Census and various datasets from Statistics Canada, including the labour force survey.

Stokes found that a total of 928,700 workers will be needed to fill job openings for occupations that usually require a university education over the decade. These positions will become vacant through retirements (and deaths) and through economic expansion. Occupations were aggregated into broad occupational categories. Below are the categories that are discussed in greater detail in the report.

- The STEM (science, technology, engineering, and mathematics) occupations will require more than 233,000 positions, including 199,000 workers and 34,000 managers.
- Occupations in education, law, and social, community, and government services, are anticipated to create a total of over 178,000 positions.
- Health occupations will require nearly 148,000 new positions, including 136,000 workers and 12,000 managers to fill job openings.
- The demand for occupations in business, finance, and administration will generate close to 128,000 positions.

Stokes also modelled the potential loss to the Ontario economy if a shortage of university educated workers was to occur. There would be repercussions across the provincial

economy, and it is estimated that for every single university educated job lost, another 0.81 jobs will be lost in the provincial economy. If the province were to experience a reduction of 10,000 university-educated workers in a single year, Ontario's GDP is estimated to fall by \$2.35 billion, and employment would fall by 18,100 jobs spanning all industries in the province. Accordingly, supporting the expansion of university graduates will be critical to Ontario's long-term economic growth and prosperity.

For more information regarding these estimates, see sections 2 and 3. The economic forecast driving these projections is presented in section 4.

2. Demand Projections

The standard method of sorting occupations in Canada is using the National Occupational Classification (NOC) system. This breaks occupations down into 500 groups of similar jobs. The occupations can then be aggregated into larger groupings for different levels of analysis. At the highest level, occupations are aggregated into five groups as follows:

1. Management occupations
2. Occupations that usually require university education – skill level A
3. Occupations that usually require college education, specialized training or apprenticeship training – skill level B
4. Occupations that usually require secondary school and/or occupation-specific training – skill level C
5. On-the-job training is usually provided for occupations – skill level D

For the purpose of this study, we have focused on occupations that usually require university education (skill level A) and the associated management occupations. The table below shows the net demand for workers from 2021 to 2030. These figures include new labour force requirements due to economic expansion, known as expansion demand, and the demand for workers to fill losses from retirements from the labour force and deaths, known as replacement demand. For those occupations that are classified in skill level A, we forecast that there will be a total of 928,700 workers required from 2021 to 2030.

Table 2.1
Net Occupational Demand for Skill Level A Workers, Ontario

	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	Total Demand 2021-2030
Total Ontario Skill Level A Occupational Demand (000s) - 122 occupations that usually require university education	153.9	105.2	100.3	92.2	77.6	74.1	78.1	81.9	84.4	81.1	928.7

Note: Excluded from these estimates are skilled positions that require a university education but are categorized under Skill level B occupations. Ontario's postsecondary education system will need to fill more than 950,000 positions in Skill level B occupations over the next 10 years. Skill level B occupations that often require a university education account for 146,000 of these positions.

The demand is highest in the first four years of the forecast. This is in part the result of the economic recovery that is expected after the pandemic and the resulting growth momentum from a rapid global growth environment. Additionally, it is important to note that university occupations fared relatively well in the pandemic and part of the relatively strong demand in the short term is due to the structural shift in the economy, such as increased use of computers and software to reduce human interactions (for instance, app development for restaurant ordering or for increased online shopping).

Table 2.2 below displays skill level A occupational demand by broad occupational category.

Table 2.2
Ontario Net Demand for University Educated Workers (Skill Level A), 2021-2030

Occupations that typically require a university degree (Skill level A)	Average Annual Demand	Total Demand 2021-2030
Management Occupations	25,454	254,541
Senior managers	6,197	61,974
Managers in engineering & computer science	3,436	34,361
Managers in healthcare	1,206	12,062
Other	14,614	146,144
Business, Finance & Administration Occupations	12,771	127,709
Finance & related	7,954	79,540
Other	4,817	48,169
STEM: Natural & Applied Sciences & Related Occupations	19,893	198,927
Engineers	4,459	44,589
Computer programmers, software engineers & designers	5,679	56,786
Other STEM	9,755	97,552
Health Occupations	13,567	135,668
Physicians	2,428	24,283
Registered nurses	7,050	70,502
Other health occupations	4,088	40,883
Occupations In Education Law & Social, Community & Government Services	17,851	178,505
Teachers (K-12)	6,734	67,338
Other	11,117	111,167
Occupations In Art Culture, Recreation & Sport	3,340	33,395
Total Demand for Occupations that Typically Require a University Degree	92,875	928,745

Governments and institutions are continuing to look towards STEM (science, technology, engineering, and mathematics) as the future of the economy. Historically, these occupations have experienced some of the strongest employment growth and relatively higher wage rates in comparison to other occupations, excluding management positions.¹ Work experience in a specific field or industry is often a requisite for management positions. Therefore, workers are typically sourced from existing workers within the STEM occupations and not from new entrants to the labour force. This process will have an additional positive impact on the demand for STEM workers as they transition to management occupations.

Over the next decade, the province will need to increase the supply of workers in STEM based occupations that usually require a university degree by an additional 198,927 workers. In addition, Ontario will require 34,361 manager positions in STEM-based fields for a total of 233,000 overall.

The demand for engineers will account for over 22% of the STEM skill A occupational demand at close to 45,000 job openings. Civil, mechanical, electrical and electronics, and computer engineers will be the highest in demand, accounting for more than 80% of total engineering occupational demand over the decade. Computer programmers, including interactive media developers and software engineers and designers, will also be in relatively high demand, representing roughly 29% of STEM skill A occupations. The former group on its own is projected to make up close to 20% of total STEM occupational demand. Other STEM occupations are comprised of 16 occupations that will be heavily impacted by occupational demand for information systems analysts and consultants, representing roughly 67% of the 97,552 jobs in this category.

The aging of Ontario's population over the decade will come with relatively strong growth in operational health spending and stimulate the demand for health workers. Total demand for university-trained health occupations is projected at 135,668 to 2030. Over 24,000 of these will be physicians, who have some of the longest training requirements relative to other occupations. The demand for registered nurses represents over half of total health job openings with just over 70,000 over the decade. The other health occupations categories will

1. Statistics Canada. No date. *Table 14-10-0340-01 Employee Wages by Occupation, Annual* (table). CANSIM (database). Last updated 27 Jan. 2021.
<https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1410034001> (accessed September 29, 2021).

see 40,000 job openings over the decade and include 13 occupations that will vary in demand. Nursing co-ordinators and supervisors, dentists, pharmacists, and physiotherapists will see the largest growth of these other occupations, accounting for more than half of the total job openings. Healthcare managers will add an additional 12,000 jobs to skill A occupational health demand, bringing the total to 148,000 jobs.

Occupations in education, law, and social, community, and government services, are anticipated to create a total of 178,505 positions over the next decade. Primary and secondary school teachers will play a big part in this figure, accounting for close to 38% of job openings with demand estimated at 67,338.

The demand for occupations in business, finance and administration will be supported by the general growth in Ontario's economy over the decade. It is estimated that the seven occupational groupings that usually require a university degree in this category will generate a total of 127,709 positions over the next 10 years. Over 60% of these job openings will be in finance-related occupations, such as financial auditors and accountants, financial and investment analysts, securities agents, investment dealers and brokers, and other financial officers.

The individual occupational demand forecast for all 122 skill level A occupations may be seen in Table 2.3 on the following pages.

Table 2.3

Net Occupational Demand – 122 Skill A Occupations

Skill Level A Occupational Demand	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	Cumulative 2021-2030
Total Ontario Occupational Demand (All 500 NOC Occupations)	512,724	415,625	427,044	358,447	269,097	240,538	250,061	261,031	267,875	245,454	3,247,896
Total Ontario Demand for Occupations That Usually Require University Education	153,853	105,161	100,331	92,215	77,563	74,055	78,095	81,914	84,418	81,140	928,745
Management Occupations That Usually Require University Education	34,294	27,743	28,904	25,999	22,402	21,574	22,591	23,463	24,178	23,393	254,541
#0011 Legislators	180	119	118	117	114	110	109	110	110	111	1,198
#0012 Senior government managers and officials	567	311	339	351	362	366	367	373	375	380	3,791
#0013 Senior managers - financial, communications and other business services	3,690	2,711	2,930	2,574	2,162	2,020	2,131	2,217	2,299	2,194	24,928
#0014 Senior managers - health, education, social & community services, membership org.	1,023	810	694	783	727	707	728	756	769	766	7,763
#0015 Senior managers - trade, broadcasting and other services, n.e.c.	1,396	1,537	1,535	1,339	1,053	991	999	1,016	1,036	977	11,879
#0016 Senior managers - construction, transportation, production and utilities	2,195	1,509	1,803	1,469	1,200	1,078	1,112	1,116	1,132	999	13,613
#0111 Financial managers	2,839	2,669	2,829	2,437	2,034	1,948	2,025	2,095	2,155	2,062	23,093
#0112 Human resources managers	1,723	1,404	1,502	1,329	1,120	1,076	1,121	1,165	1,206	1,163	12,809
#0113 Purchasing managers	830	644	706	606	521	510	520	528	539	507	5,911
#0114 Other administrative services managers	1,140	914	958	877	747	718	747	775	799	773	8,448
#0121 Insurance, real estate and financial brokerage managers	1,171	1,097	1,142	949	856	826	859	888	908	873	9,569
#0122 Banking, credit and other investment managers	1,469	2,751	2,763	2,333	1,810	1,733	1,864	1,958	2,015	1,937	20,633
#0124 Advertising, marketing and public relations managers	2,699	2,022	2,222	1,781	1,407	1,357	1,450	1,541	1,635	1,559	17,673
#0125 Other business services managers	785	711	770	633	498	480	503	521	541	514	5,956
#0131 Telecommunication carriers managers	312	138	327	236	311	283	292	302	317	315	2,833
#0132 Postal and courier services managers	137	154	180	142	89	84	96	101	101	94	1,178
#0211 Engineering managers	886	564	725	602	515	505	504	507	523	492	5,823
#0212 Architecture and science managers	426	233	271	249	219	220	226	232	239	231	2,546
#0213 Computer and information systems managers	4,349	2,873	3,032	2,893	2,502	2,307	2,516	2,645	2,759	2,662	28,538
#0311 Managers in health care	1,634	1,160	897	1,128	1,126	1,144	1,177	1,245	1,266	1,285	12,062
#0411 Government managers - health and social policy development & program admin.	246	117	134	143	154	159	161	165	166	170	1,615
#0412 Government managers - economic analysis, policy development & program admin.	303	143	165	174	184	190	193	197	200	204	1,953
#0413 Government managers - education policy development and program admin	33	16	18	19	20	20	21	22	22	23	214
#0414 Other managers in public administration	258	127	145	154	163	167	169	174	176	180	1,713
#0421 Administrators - post-secondary education and vocational training	736	455	239	230	207	197	237	278	303	316	3,198
#0422 School principals and administrators of elementary and secondary education	919	741	606	612	640	691	738	757	768	778	7,250
#0423 Managers in social, community and correctional services	867	742	709	749	714	711	726	751	769	772	7,510
#0431 Commissioned police officers	45	26	28	30	30	30	30	31	31	31	312
#0432 Fire chiefs and senior firefighting officers	59	41	43	44	44	42	42	42	43	43	443
#0433 Commissioned officers of the Canadian Forces	615	234	293	319	354	383	390	399	404	418	3,809
#0511 Library, archive, museum and art gallery managers	86	120	133	121	90	85	90	93	94	91	1,003
#0512 Managers - publishing, motion pictures, broadcasting and performing arts	224	225	239	201	192	179	181	186	191	189	2,007
#0513 Recreation, sports and fitness program and service directors	451	422	408	375	239	255	266	278	289	285	3,268
Total Non Management Occupations That Usually Require University Education	119,559	77,418	71,427	66,216	55,161	52,481	55,504	58,451	60,240	57,747	674,204
Business, Finance & Administration Occupations	19,495	16,979	17,582	13,908	10,246	9,471	9,772	10,143	10,484	9,629	127,709
#1111 Financial auditors and accountants	6,241	4,593	4,974	3,512	2,556	2,393	2,387	2,450	2,547	2,288	33,941
#1112 Financial and investment analysts	1,399	1,778	1,788	1,436	1,053	976	1,044	1,098	1,132	1,055	12,759
#1113 Securities agents, investment dealers and brokers	354	681	661	528	378	344	364	376	380	349	4,415
#1114 Other financial officers	2,743	4,215	4,135	3,338	2,429	2,217	2,323	2,391	2,415	2,219	28,425
#1121 Human resources professionals	2,154	1,547	1,590	1,403	1,125	1,058	1,080	1,117	1,147	1,081	13,302
#1122 Professional occupations in business management consulting	3,676	2,172	2,324	2,055	1,548	1,408	1,434	1,483	1,549	1,415	19,064
#1123 Professional occupations in advertising, marketing and public relations	2,927	1,993	2,111	1,636	1,157	1,077	1,140	1,229	1,315	1,222	15,807
Natural & Applied Sciences & Related Occupations	38,519	20,561	23,759	20,974	16,640	14,582	15,602	16,142	16,789	15,359	198,927
#2111 Physicists and astronomers	117	60	60	53	46	47	48	49	50	48	578
#2112 Chemists	669	203	274	246	204	211	211	204	200	175	2,597
#2113 Geoscientists and oceanographers	166	67	95	88	70	59	54	59	65	60	783
#2114 Meteorologists and climatologists	23	9	10	9	10	11	10	10	10	10	112
#2115 Other professional occupations in physical sciences	47	29	32	25	21	19	17	17	18	16	241
#2121 Biologists and related scientists	599	219	251	239	197	203	206	218	228	218	2,578
#2122 Forestry professionals	40	19	18	15	15	14	14	16	16	14	181
#2123 Agricultural representatives, consultants and specialists	64	48	41	36	26	25	25	25	26	22	338
#2131 Civil engineers	2,022	633	1,421	1,092	784	711	696	705	771	698	9,533
#2132 Mechanical engineers	1,867	1,133	1,596	1,156	875	821	753	729	762	661	10,353
#2133 Electrical and electronics engineers	1,960	988	1,341	1,028	844	861	857	819	819	718	10,235
#2134 Chemical engineers	400	147	216	181	141	138	135	131	132	116	1,737
#2141 Industrial and manufacturing engineers	573	421	498	350	267	259	239	225	228	190	3,250
#2142 Metallurgical and materials engineers	69	50	59	43	32	30	27	27	28	24	389
#2143 Mining engineers	64	38	45	36	33	26	19	21	24	22	328
#2144 Geological engineers	76	12	48	36	26	25	24	25	28	25	325
#2145 Petroleum engineers	17	9	15	13	9	7	8	8	9	8	103
#2146 Aerospace engineers	128	210	207	120	82	66	52	49	56	45	1,015
#2147 Computer engineers (except software engineers and designers)	1,217	655	769	655	561	509	555	566	583	533	6,603
#2148 Other professional engineers, n.e.c.	149	70	87	72	56	57	57	57	59	54	718

#2151 Architects	824	167	531	399	298	289	273	285	308	279	3,653
#2152 Landscape architects	93	30	65	50	38	37	35	36	38	35	457
#2153 Urban and land use planners	422	169	263	219	174	164	165	170	179	171	2,096
#2154 Land surveyors	212	59	155	119	85	77	74	73	79	72	1,005
#2161 Mathematicians, statisticians and actuaries	443	303	314	272	220	213	222	233	242	231	2,693
#2171 Information systems analysts and consultants	12,037	7,129	7,337	6,912	5,564	4,741	5,205	5,443	5,619	5,198	65,185
#2172 Database analysts and data administrators	1,130	739	759	694	566	499	533	554	568	529	6,571
#2173 Software engineers and designers	3,974	2,136	2,262	2,092	1,658	1,392	1,576	1,654	1,728	1,579	20,051
#2174 Computer programmers and interactive media developers	7,319	3,910	4,054	3,843	3,062	2,530	2,876	3,041	3,176	2,924	36,735
#2175 Web designers and developers	1,802	900	937	882	674	541	637	693	739	683	8,488
Health Occupations	22,303	14,110	9,695	12,866	12,290	12,208	12,451	13,261	13,278	13,206	135,668
#3011 Nursing co-ordinators and supervisors	757	521	387	501	499	507	524	557	567	576	5,396
#3012 Registered nurses and registered psychiatric nurses	11,529	7,166	4,736	6,621	6,527	6,594	6,576	6,932	6,916	6,905	70,502
#3111 Specialist physicians	1,891	1,157	729	1,040	1,004	988	1,011	1,079	1,077	1,075	11,051
#3112 General practitioners and family physicians	2,419	1,467	875	1,222	1,153	1,110	1,168	1,268	1,275	1,275	13,232
#3113 Dentists	986	626	405	555	522	493	520	557	553	549	5,766
#3114 Veterinarians	335	223	250	127	71	73	70	76	86	74	1,385
#3121 Optometrists	258	171	118	154	145	137	144	154	153	152	1,586
#3122 Chiropractors	436	292	203	265	250	239	251	267	266	264	2,733
#3124 Allied primary health practitioners	449	283	186	259	252	249	257	275	275	275	2,760
#3125 Other professional occupations in health diagnosing and treating	226	147	102	135	128	123	130	139	139	138	1,407
#3131 Pharmacists	630	604	764	649	453	412	450	496	493	437	5,388
#3132 Dietitians and nutritionists	400	239	170	226	210	209	216	233	235	234	2,372
#3141 Audiologists and speech-language pathologists	318	202	134	181	178	181	192	205	206	206	2,003
#3142 Physiotherapists	854	520	326	478	463	456	481	521	526	530	5,155
#3143 Occupational therapists	456	280	179	264	260	261	274	297	301	304	2,876
#3144 Other professional occupations in therapy and assessment	358	214	132	190	176	177	187	205	209	212	2,060
Occupations In Education Law & Social, Community & Government Services	33,925	21,479	16,058	14,750	13,365	13,768	15,114	16,241	16,918	16,887	178,505
#4011 University professors and lecturers	2,729	1,559	419	304	191	137	309	506	618	671	7,443
#4012 Post-secondary teaching and research assistants	2,020	834	-252	-327	-411	-431	-221	12	163	252	1,639
#4021 College and other vocational instructors	2,895	1,683	992	949	799	723	849	965	1,028	1,034	11,917
#4031 Secondary school teachers	3,040	2,689	1,856	2,015	2,006	2,023	1,970	1,974	1,929	1,951	21,453
#4032 Elementary school and kindergarten teachers	7,024	4,426	3,047	2,989	3,443	4,197	4,931	5,155	5,293	5,380	45,885
#4033 Educational counsellors	592	354	176	167	152	153	187	217	234	241	2,473
#4111 Judges	105	50	51	51	52	50	49	50	49	49	556
#4112 Lawyers and Quebec notaries	2,967	1,882	2,155	1,089	585	581	563	619	715	607	11,763
#4151 Psychologists	804	521	362	449	423	407	421	444	443	440	4,714
#4152 Social workers	1,456	1,107	927	1,069	1,048	1,047	1,063	1,106	1,124	1,130	11,077
#4153 Family, marriage and other related counsellors	635	450	348	416	395	387	393	411	415	414	4,264
#4154 Professional occupations in religion	831	814	779	836	646	566	559	566	564	519	6,680
#4155 Probation and parole officers and related occupations	158	55	63	69	75	73	74	78	79	80	804
#4156 Employment counsellors	319	282	256	235	203	197	205	212	217	213	2,339
#4161 Natural and applied science policy researchers, consultants and program officers	887	376	445	416	369	366	362	373	384	372	4,350
#4162 Economists and economic policy researchers and analysts	662	521	551	479	400	389	402	415	422	406	4,647
#4163 Business development officers and marketing researchers and consultants	2,523	1,692	1,860	1,438	1,028	938	984	1,034	1,092	987	13,576
#4164 Social policy researchers, consultants and program officers	1,103	520	579	561	540	542	544	558	566	563	6,076
#4165 Health policy researchers, consultants and program officers	1,391	763	639	715	659	656	671	711	725	717	7,647
#4166 Education policy researchers, consultants and program officers	688	359	240	260	240	235	267	293	307	312	3,201
#4167 Recreation, sports and fitness policy researchers, consultants and program officers	261	183	167	168	128	127	131	138	143	143	1,589
#4168 Program officers unique to government	614	239	279	291	307	315	311	312	308	311	3,287
#4169 Other professional occupations in social science, n.e.c.	218	121	119	110	89	87	89	94	97	94	1,118
Occupations In Art Culture , Recreation & Sport	5,317	4,289	4,333	3,718	2,620	2,452	2,565	2,664	2,771	2,666	33,395
#5111 Librarians	78	146	148	133	135	104	105	103	102	96	1,150
#5112 Conservators and curators	54	67	74	64	38	36	38	39	39	36	485
#5113 Archivists	27	27	27	24	20	18	18	19	19	18	217
#5121 Authors and writers	1,057	896	935	767	484	468	469	479	493	452	6,500
#5122 Editors	411	392	422	354	317	293	302	314	328	321	3,454
#5123 Journalists	174	174	201	158	142	131	121	121	126	121	1,469
#5125 Translators, terminologists and interpreters	443	310	327	218	158	153	148	151	157	141	2,206
#5131 Producers, directors, choreographers and related occupations	643	685	679	493	455	386	412	432	454	448	5,087
#5132 Conductors, composers and arrangers	133	141	136	119	72	68	68	69	70	65	941
#5133 Musicians and singers	1,137	538	489	578	369	365	421	448	470	476	5,291
#5134 Dancers	247	4	-10	46	26	26	54	67	79	91	630
#5135 Actors and comedians	281	316	305	253	123	125	131	138	145	137	1,954
#5136 Painters, sculptors and other visual artists	632	592	602	511	281	280	279	285	289	264	4,015

3. The Economic Contribution of University Educated Workers

University-educated workers contribute to a significant portion of Ontario's economic output. In this section of the study, the estimated impact on GDP and employment if a portion of the future demand for skill type A workers were to go unfilled is presented.

The approach adopted to assess the economic impacts of a shortage of university-trained workers in the Ontario economy is to conduct a “base case” scenario where those workers are present, and then compare the results to a scenario where there is a shortage of workers, referred to as a “shock” to the model.

A shock was carried out for the Ontario economy where it was assumed that there is a shortage of 10,000 workers in 2022. This was done first at the occupational level, where it was assumed that the reduction in skill type A occupations was based on the current share of these jobs in the economy. Therefore, job losses were evenly spread across all occupations that usually required a university degree. The reductions in occupations were then mapped backwards through the occupational model to the industries in which these occupations are employed. This method results in an estimate of reductions in industry employment. Average industry productivity was then applied to these employment losses to estimate direct output losses to the economy.

A shortage of university-educated workers will have direct, indirect, and induced impacts on the economy. The direct impacts are those that are directly associated with the reduction. In this example, fewer workers and less production is associated with this loss. The indirect impacts are the result of fewer inputs being purchased and sold between firms as they scale back production due to reduced output. Finally, the induced impacts are the result of less income and profits from production which in turn results in fewer purchases by households in the economy.

The summation of the direct, indirect, and induced impacts relative to the direct impacts is known as a Type II multiplier. The employment and GDP multipliers were estimated at 1.8 and 2.35, respectively, as shown in Table 3.1 below. This means that for every single job lost,

another 0.81 was lost in the economy. For every \$1 direct reduction in GDP another \$1.35 was reduced from indirect and induced sources. Total impacts in 2022 for a reduction of 10,000 university-trained workers is \$2.35 billion, with a total employment loss of 18,100 jobs across the Ontario economy.

Table 3.1
Impact of a reduction in 10,000 university educated workers in a single year

	Direct Impact	Total Impact	Type II Multiplier
Employment (000s)	-10.0	-18.1	1.81
GDP (\$2012 Millions)	-995	-2,341	2.35

The higher impacts on GDP can be explained by a number of factors. Many of the jobs lost are in relatively high-productivity industries and in occupations that pay above the average. This will have a disproportionately larger impact on output due to the high productivity factor, and as household income is reduced more so than in relatively lower paying occupations.

It is important to note that the economic loss scenario is a static one-year impact analysis. Longer term effects of continued loss in educated and skilled individuals are not considered due to the complexities in calculating the appropriate externalities. The loss of innovation is hard to predict or measure.

The loss of university-educated individuals will have an impact on the education of future generations as the children of lower-educated parents are less likely to undertake further education. A continued shortage of skilled individuals may cause potential investors to instead look outside of Ontario, further affecting the economy over time.

4. Ontario Economic Outlook

The Stokes Economics July 2021 forecast provides the economic backdrop for the employment and occupation forecasts used in this report. The near-term outlook for the Ontario economy shows a strong recovery. Real GDP is expected to increase by 5.5% in 2021, with an employment rebound of 4.5%. Industries that were the hardest hit by the pandemic will take a few years to fully recover. The longer-term forecast horizon to 2030 is relevant for this study, and Ontario's real GDP growth is expected to average 2.8% annually. A series of key economic indicators are presented in Table 4.1 on the following pages.

The economic forecast was produced in June of 2021. More recent information suggests that supply chain issues and inflationary pressures will impact the Ontario outlook in the short term.

Employment Outlook

The Ontario macroeconomic forecast drives the province's industry employment. The economy is experiencing a snap back in demand as the economy recovers from the pandemic. Rapid growth in the global economy, supported by government stimulus measures in Canada and internationally will significantly boost employment growth in university-educated occupations. The following provides recent employment levels for a number of industries that employ large numbers of university graduates:

- Professional, Scientific and Technical Services industry employed 665,000 workers last year and is expected to expand by 165,000 over the coming decade. Computer Systems Design and Related Services is the largest subindustry employer and had 241,000 workers last year and is expected to add 104,000 over the next 10 years. Twenty-two percent of the jobs in occupations that usually require a university degree are in the Professional Scientific and Technical industry – currently 457,000 workers.
- Management of Companies and Enterprises, and Administrative and Support Services industry employed 295,000 workers last year. Employment is expected to rise by about 40,000 over the next 10 years.
- Heavy and Civil Engineering Construction employed 44,000 in 2020. Employment will be flat in this industry over the coming decade.

- Finance, Insurance, Real Estate, Rental and Leasing services industry employed 607,000 workers last year and is expected to increase by a further 121,000 over the coming decade. Finance is the largest subindustry and employed 333,000. Insurance carriers employed 122,000. Thirteen percent of the jobs in occupations that usually require a university degree are in the Finance, Insurance and Real Estate industry – currently 273,000 workers.
- Educational Services employed 516,000 workers last year and is expected to increase by a rather modest 27,000 over the next 10 years. Elementary and secondary schools employed 305,000 workers. Seventeen percent of jobs in the occupations that usually require a university degree are in the Education Services industry – currently 353,000 workers.
- Health Care and Social Assistance industry employed 870,000 workers last year and is projected to increase by 240,000 over the coming decade. Hospitals employed 266,000 last year. Nursing and Residential Care Facilities employed 172,000 last year and employment will rise by 50,000 over the coming decade. Seventeen percent of jobs in the occupations that usually require a university degree are in the Health Care and Social Assistance industry – currently 352,000 workers.

Demographic Outlook

The Ontario population has been growing rapidly in recent years, largely due to increased international migration. The number of non-permanent residents has been growing at a rapid rate, mainly as a result of increases in international students coming to the province. In addition, the number of immigrants has increased significantly, with federal targets looking towards immigration to boost economic growth and fill gaps in the labour market that has been facing significant pressures from both an aging population and increased demand for workers due to economic expansion.

The Ontario population has been aging with the impacts of the post-war baby boom and the subsequent baby bust. This created a significant population in the age range in which retirement rates increase for both men and women. The result of the aging population will be an increased loss in experienced and skilled workers, as the level of retirements increase. Job openings from retirements will continue to increase over the forecast period, creating significant demands on training and education for new workers.

Table 4.1
Key Economic Indicators, Ontario

	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	Average 2021-25	Average 2026-30
Real GDP (\$2012 Millions)	752,295	793,943	833,595	859,163	881,455	899,020	914,601	931,729	950,569	970,229	986,074		
% Change	-5.8	5.5	5.0	3.1	2.6	2.0	1.7	1.9	2.0	2.1	1.6	3.6	1.9
Consumer Expenditures	435,912	463,345	489,434	514,488	535,492	550,504	563,966	577,931	594,035	610,419	624,553		
% Change	-6.1	6.3	5.6	5.1	4.1	2.8	2.4	2.5	2.8	2.8	2.3	4.8	2.6
Residential Investment	60,492	66,955	67,495	68,320	68,444	68,495	66,975	67,426	68,635	69,192	68,579		
% Change	7.8	10.7	0.8	1.2	0.2	0.1	-2.2	0.7	1.8	0.8	-0.9	2.6	0.0
Plant & Equipment Investment	45,980	51,561	58,602	62,382	66,038	67,446	67,510	67,264	67,091	69,004	69,096		
% Change	-14.6	12.1	13.7	6.5	5.9	2.1	0.1	-0.4	-0.3	2.9	0.1	8.1	0.5
Government Investment	29,204	29,737	30,701	31,907	32,946	32,009	31,899	31,849	32,174	32,353	32,500		
% Change	0.5	1.8	3.2	3.9	3.3	-2.8	-0.3	-0.2	1.0	0.6	0.5	1.9	0.3
Government Current Expenditures	156,803	157,691	158,661	160,114	162,172	164,405	166,674	169,092	171,729	174,427	177,239		
% Change	-1.7	0.6	0.6	0.9	1.3	1.4	1.4	1.5	1.6	1.6	1.6	1.0	1.5
Exports	363,984	386,391	410,936	425,212	434,893	445,352	453,885	460,774	468,582	476,510	485,261		
% Change	-8.7	6.2	6.4	3.5	2.3	2.4	1.9	1.5	1.7	1.7	1.8	4.2	1.7
Imports	365,149	399,973	427,616	447,348	465,952	477,864	484,859	492,140	503,166	515,010	525,403		
% Change	-9.2	9.5	6.9	4.6	4.2	2.6	1.5	1.5	2.2	2.4	2.0	5.6	1.9
Source Population (000s)	12,218	12,326	12,502	12,673	12,832	12,984	13,126	13,262	13,393	13,530	13,666		
% Change	1.4	0.9	1.4	1.4	1.3	1.2	1.1	1.0	1.0	1.0	1.0	1.2	1.0
Labour Force (000s)	7,766	7,989	8,141	8,278	8,388	8,469	8,533	8,591	8,640	8,698	8,756		
% Change	-0.6	2.9	1.9	1.7	1.3	1.0	0.8	0.7	0.6	0.7	0.7	1.8	0.7
Employment (000s)	7,022	7,334	7,549	7,773	7,931	8,002	8,046	8,099	8,163	8,234	8,283		
% Change	-4.8	4.5	2.9	3.0	2.0	0.9	0.6	0.7	0.8	0.9	0.6	2.7	0.7
Participation Rate (%)	63.6	64.8	65.1	65.3	65.4	65.2	65.0	64.8	64.5	64.3	64.1	65.2	64.5
Unemployment Rate (%)	9.6	8.2	7.3	6.1	5.5	5.5	5.7	5.7	5.5	5.3	5.4	6.5	5.5
Hourly Labour Productivity (\$2012)	65.2	64.6	64.8	64.7	65.0	65.9	66.9	67.7	68.6	69.4	70.1		
% Change	3.4	-0.9	0.3	-0.2	0.4	1.5	1.4	1.3	1.2	1.2	1.1	0.2	1.2
Real GDP Per Capita (\$2012 000s)	51.1	53.6	55.5	56.5	57.3	57.8	58.1	58.6	59.2	59.8	60.2		
% Change	-7.0	5.0	3.6	1.8	1.4	0.8	0.6	0.8	1.0	1.0	0.6	2.5	0.8
Population (000s)	14,734	14,815	15,013	15,204	15,386	15,563	15,733	15,895	16,054	16,221	16,389		
% Change	1.3	0.5	1.3	1.3	1.2	1.2	1.1	1.0	1.0	1.0	1.0	1.1	1.0
Net In-Migration (000s)	161.7	49.6	168.3	159.7	149.0	144.5	136.3	130.1	128.0	138.4	141.5	134.2	134.9
Net Interprovincial Migration	0.4	-2.8	3.9	1.7	-5.1	-4.7	-12.0	-17.1	-18.9	-9.0	-4.9	-1.4	-12.3
Net International Migration	161.3	52.4	164.4	158.0	154.0	149.2	148.3	147.1	146.9	147.4	146.4	135.6	147.2
Households (000s)	5,697	5,764	5,868	5,969	6,067	6,164	6,255	6,337	6,415	6,495	6,576	5,967	6,416
Household Formation (000s)	91.6	67.5	104.0	101.1	97.8	97.3	90.8	82.1	77.8	80.4	80.5	93.5	82.3
Housing Starts (000s)	81.3	94.5	84.6	82.6	82.9	81.3	79.8	81.3	82.7	82.5	80.2	85.2	81.3
Single Detached	22.7	30.9	27.8	27.0	25.8	26.5	23.7	24.1	25.0	25.3	24.5	27.6	24.5
Multi-family	58.6	63.6	56.8	55.6	57.1	54.8	56.2	57.2	57.7	57.2	55.7	57.6	56.8
GDP Deflator (2012=1)	1.124	1.166	1.185	1.207	1.228	1.250	1.273	1.297	1.321	1.347	1.373		
% Change	0.6	3.7	1.7	1.8	1.8	1.8	1.9	1.9	1.8	1.9	1.9	2.2	1.9
Consumer Price Index (2002=1)	1.384	1.422	1.454	1.480	1.500	1.527	1.555	1.586	1.618	1.652	1.687		
% Change	0.6	2.8	2.2	1.8	1.8	1.8	1.9	2.0	2.0	2.1	2.1	2.1	2.0
Labour Income Per Hour (\$)	40.6	41.7	42.8	44.7	46.8	48.9	51.1	53.4	55.9	58.4	61.0		
% Change	8.3	2.8	2.7	4.4	4.6	4.5	4.5	4.5	4.7	4.6	4.4	3.8	4.5
Retail Sales (\$Millions)	218,650	236,845	253,021	272,132	291,630	308,036	323,705	340,621	359,908	380,766	400,269		
% Change	-5.5	8.3	6.8	7.6	7.2	5.6	5.1	5.2	5.7	5.8	5.1	7.1	5.4
Personal Income (\$Millions)	766,311	793,676	837,325	891,126	949,023	1,002,019	1,057,017	1,115,958	1,180,153	1,248,327	1,314,998		
% Change	6.0	3.6	5.5	6.4	6.5	5.6	5.5	5.6	5.8	5.8	5.3	5.5	5.6
Net Operating Surplus Corporations (\$M)	95,265	88,746	97,164	88,764	82,728	84,546	84,915	84,764	83,272	82,511	79,417		
% Change	-15.0	-6.8	9.5	-8.6	-6.8	2.2	0.4	-0.2	-1.8	-0.9	-3.8	-2.1	-1.3

Source: Statistics Canada and Stokes Economics

5. Conclusion

The Ontario economy is expected to experience a tightening in its labour market as it recovers from the COVID-19 pandemic and remain relatively tight over the remainder of the decade. Population aging, increasing retirements, and economic expansion will result in a requirement of an additional 928,700 university-educated workers.

Ontario universities will need capacity to meet the demand for these highly skilled workers.

Appendix



Stokes Economics is Canada's leading detailed occupational labour market information provider, macroeconomic model builder, and forecast data centre. Stokes maintains national, provincial and regional economic models that are used to develop economic outlooks and customized impact studies ranging from economic and fiscal projections to occupational human resource planning. The models are updated semi-annually and calibrated to the latest economic and demographic data available. Stokes also closely tracks major construction projects for a more accurate estimate of investment and economic growth.

Macroeconomic Model - Ontario

The projections are prepared using the Stokes Economics multi-sector macroeconomic model of the Ontario economy.

The Ontario macroeconomic model is large and covers over 4,000 variables – more than 1,500 of these are endogenous variables, or are determined within the model. Over 50 industries are modelled with about two thirds of the sector detail focused on the goods producing industries that includes a strong level of detail in the mining and manufacturing sectors. The aggregate demand side of the model is supplemented by an extensive inventory of major capital projects, which is a key driver of the provincial economy. These are KLEM models in which capital (K), labour (L), energy (E) and non-energy materials (M) are combined to produce gross output for each industry. They rely heavily on input-output table information from Statistics Canada to inform inter-industry and intra-industry supply and demand relationships (i.e. total demand by each sector from all other sectors and from an industries' own sector).

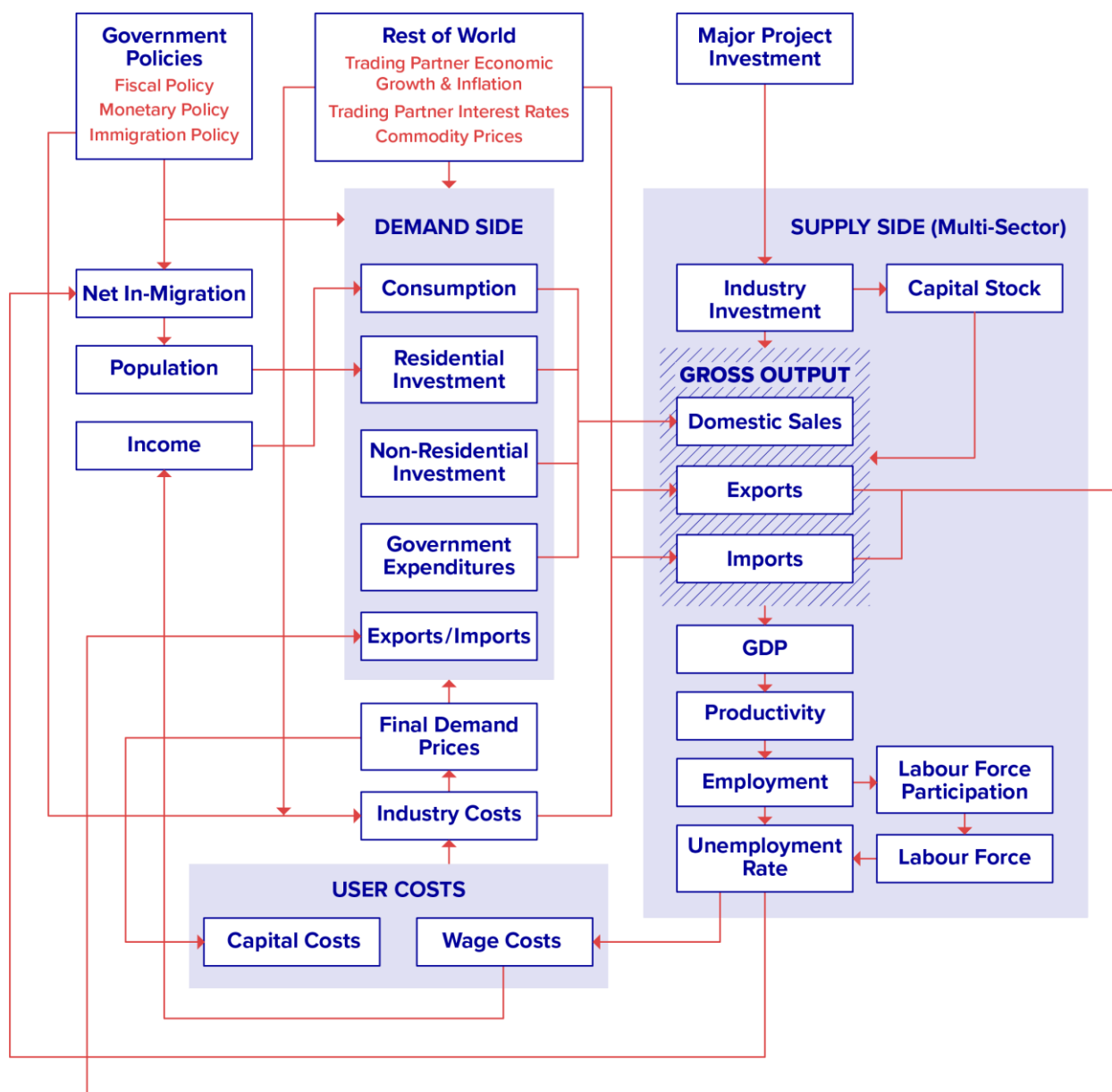
Population is an endogenous variable in the model and very detailed demographic projections are produced using a cohort-survival projection method. This approach forecasts what the future population will be based on the survival of the existing population and the

births that will occur. Stokes supplements this approach with interprovincial and international migration that responds to economic opportunities and adjusts to allow for labour market balancing – known as optimal immigration. International migration is also supplemented with national immigration policies. Both interprovincial and international migration account for the age structure of migrants. Population growth and the age structure of the population guide the demand for housing as the model estimates housing starts by structure type. This approach to demographic projections is employed in all Stokes' provincial and regional economic models.

The models employ a stage of process pricing approach in which raw material prices are linked to producer prices and final demand prices through input-output relationships. This cost-push approach allows the model to capture the pass-through of changes in energy and non-energy commodity prices to the cost structure of industries. Industry cost and price relationships inform the model's trade sectors to determine how local costs compare against industry prices to estimate changes in industry exports and imports. The industry cost model aids in modelling the cost of capital relative to labour and energy, which allows for substitution between the factors of production. This approach generates economic cycles as relative prices between the factors of production change.

While the model involves the simultaneous decisions of various factors, its basic workings can be seen in Figure A.1 on the next page.

Figure D.1: Macroeconomic Model



Provincial Occupation Modelling System (POMS)

The Stokes POMS produces information that can be used to help Labour Market Information (LMI) analysts and decision makers identify significant supply-demand gaps across the occupations and the possible sources of supply to remove these gaps. The analysis presented in this paper focused solely on the use of the demand side results of the POMS model, as supply was estimated using an alternative methodology that incorporated historical information provided by the Council of Ontario Universities (COU). A description of the POMS methodology regarding estimating demand is described below.

The Employment and Social Development Canada (ESDC) National Occupational Classification (NOC) system of occupation definitions is adopted for the modelling system – <https://noc.esdc.gc.ca/>. There are 4 levels of NOC occupations. An example of these levels is:

- 0 Management occupations – 1 digit
 - 00 Senior management occupations – 2 digit
 - 001 Legislators and senior management – 3 digit
 - 0011 Legislators – 4 digit

The occupational modelling system works at the 4-digit level of occupation aggregation where there are 500 occupations.

The occupation concepts described above are measured on a **Place of Residence** basis and refer to persons living in a province. Persons living in a province may have a **Place of Work** in the province or outside the province. The occupation modelling system focuses on the demand for workers in the province. To the extent that occupations work largely outside the province or are dominated by workers living outside the province, it will present a less accurate picture of **Place of Residence** measures for the province.

The data for the modelling system are sourced from the Census of Canada and the Labour Force Survey (LFS). They are measured on an LFS basis for the occupations as a whole. That is, the total labour force and total employment are equal to the LFS values each year. Nevertheless, the occupation data do not match the LFS occupation data published by

Statistics Canada. Statistics Canada occupation data generally has too small a sample size to provide reliable estimates of the occupations, particularly on a provincial basis.

To estimate occupation demand for a given occupation, an occupation's share of Census employment of Census total employment in an industry is applied to the corresponding LFS industry employment. Labour force demand for a specific occupation is then estimated in a two-step procedure. In the first step the ratio of Census labour force to Census employment share is applied to the LFS equivalent employment created above. Next, the resulting labour force estimates are normalized to total LFS labour force. The employment share coefficients for the occupation employment part of the model were estimated as the ratio of an occupation's employment in an industry to the industry's total employment for the latest Census year.

The occupation modelling system on the demand side focuses on the workforce required by private and public organizations in the economy. The total occupational demand requirements are changes in employment, people retiring from the workforce, and people dying.

The key demand side concepts for an occupation are as follows:

- **Job Openings** (flow): the summation of Expansion Demand and Replacement Demand;
- **Expansion Demand** (flow): the change in labour force demand (stock) as defined above, in a given year;
- **Replacement Demand** (flow): the number of retirements (flow) and deaths (flow) from an occupation in a given year;

The above concepts are of interest in describing the sources of change in the demand for an occupation. These concepts similarly extend to the demand for a group of occupations – in the case of this study, university educated occupations – by summing these occupational demand components.